

Theory

Abstract

We define mechanically checkable admissibility and comparison structure for regret evaluation in heterogeneous agent swarms under deterministic replay. Evaluation is performed per task context, within which metric definitions, units, scales, and ordering relations are strictly frozen. Deterministic trace invariance and causal closure enable counterfactual comparison of admissible alternatives from identical boundary conditions. Regret is defined as a context-relative, partially ordered quantity, valid only where unit consistency and admissible comparators are satisfied. This specification is evaluation-only: it detects coordination failures and misattribution under replay and is independent of learning or optimization dynamics.

1 Governing Definitions and Scope

This document specifies *necessary* conditions for mechanically valid regret evaluation and attribution in heterogeneous agent swarms. The scope is restricted to evaluation under deterministic replay and counterfactual comparison within a fixed task context. Learning, adaptation, and policy synthesis are explicitly out of scope; regret in this document is *not* a learning signal. Design intent: evaluation invariants must precede learning so that measured changes are distinguishable from measurement drift, comparator invalidity, or oracle leakage.

Task context and frozen evaluation structure. Evaluation is conducted relative to a task context τ . A task context specifies the metric set, units, dimensions, scales (including normalization constants), weights (if used), and ordering relations used for regret attribution. Once τ is fixed, all metric definitions and ordering relations are immutable for the duration of evaluation. Comparisons across distinct task contexts are undefined because τ induces the comparison domain.

System closure and deterministic trace invariance. For a fixed task context τ , the swarm and its environment are assumed to form a causally closed, replayable system. For any boundary condition s and agent alternative a , the execution trace and all derived metrics are deterministic. Deterministic trace invariance holds if and only if identical boundary conditions and alternatives yield identical traces and metric vectors under replay. Note: this is logical determinism of the evaluation function. Hardware or scheduler nondeterminism is permitted only if it is elevated into state s (e.g., recorded schedules or captured RNG streams) so that replay remains a pure function.

Let M_τ denote the task-context-specific multi-metric observation map,

$$M_\tau : (s, a) \mapsto M_\tau(s, a), \tag{1}$$

where $M_\tau(s, a)$ is a vector of metrics with fixed units and dimensions determined by τ . Replay determinism requires M_τ to be a function.

Mechanical admissibility of counterfactuals. Counterfactual comparison is admissible only when mechanically verifiable conditions are satisfied. For a fixed task context τ and boundary condition s , two alternatives a and a' are admissible for comparison if and only if evaluation begins from identical s and deterministic trace invariance holds.

We write $(s, a) \sim_\tau (s, a')$ to denote admissible comparability under τ . Admissibility is a decidable property of the evaluation setup, not a modeling judgment. An inadmissible comparison is an *invalid counterfactual* and must fail closed: evaluation terminates without emitting regret values.

Heterogeneity and attribution. Agents in the swarm may be heterogeneous in structure, capability, and internal state. Regret attribution is defined over alternatives rather than agent identities. Attribution is valid only when admissibility holds under τ , independent of agent heterogeneity.

Ordering and regret. For a fixed task context τ , the metric vectors induced by M_τ admit a partial order $\leq_\tau$ sufficient for regret attribution (total orders are a special case). Regret is defined only for admissible comparisons and must respect the frozen comparison structure, unit consistency, and causal closure imposed by τ . Comparisons that violate these constraints are undefined.

Scope (hard requirements). The following requirements are *mandatory for validity*. Violations invalidate regret attribution and must result in a termination of the evaluation sequence without emitting results.

- **Offline measurement (validity):** Metric computation and regret attribution are performed strictly post-execution. No evaluation feedback (gradients, reward signals) is permitted during execution.
- **Replayability (validity):** For a fixed (`spec`, `seed`, `trace`), execution is a pure function $M_\tau(s, a)$ and must be reproducible bit-for-bit under deterministic replay.
- **No partial resets (validity):** If any external dependency (e.g., API state, disk I/O) cannot be reset to the initial boundary condition s , the counterfactual is undefined and the system must fail closed.
- **No online adaptation (validity):** The agent alternative a (including policies, prompts, and tool access) remains immutable for the duration of the evaluation run to prevent policy drift.
- **Unit consistency (validity):** Any aggregate regret scalar over multiple metrics must be dimensionless via explicit normalization (r_i). Each metric must declare physical units and a characteristic scale $C_i(s)$ prior to aggregation.

Out of scope. Algorithmic design choices (search, learning, and control strategies) are not prescribed. This document defines the invariants required for *valid measurement*, not the mechanisms for *performance optimization*.

2 Problem Formulation

We model evaluation as a *closed, replayable system* operating under a fixed task context τ . The system is defined by the following components:

- **State s :** Complete boundary conditions for evaluation. This includes immutable workload traces, environment reset state, RNG seed state, and all other factors that causally influence state transitions, observations, or metric realizations.

- **Alternative** $a \in A$: A finite, enumerated set of admissible actions, policies, or designs, enumerated prior to execution for a given evaluation run. An alternative is admissible only if its execution depends exclusively on its declared information set $I(h)$ and does not access undeclared state, oracle information, or external side channels.
- **Outcome** $M_\tau(s, a)$: A deterministic mapping from (s, a) to observed metrics, realized by offline analysis of execution artifacts. Replay determinism requires that M_τ is a pure function.

2.1 Closed-System Assumption (Causal Closure)

All counterfactual evaluations are performed against the same boundary conditions s . If any external dependency or environmental influence cannot be reset, mocked, or deterministically replayed as part of the state s , the counterfactual is undefined. In such cases, the evaluation *must terminate* without emitting regret values to prevent the attribution of entropy as regret.

2.2 Evaluation Pipeline (Overview)

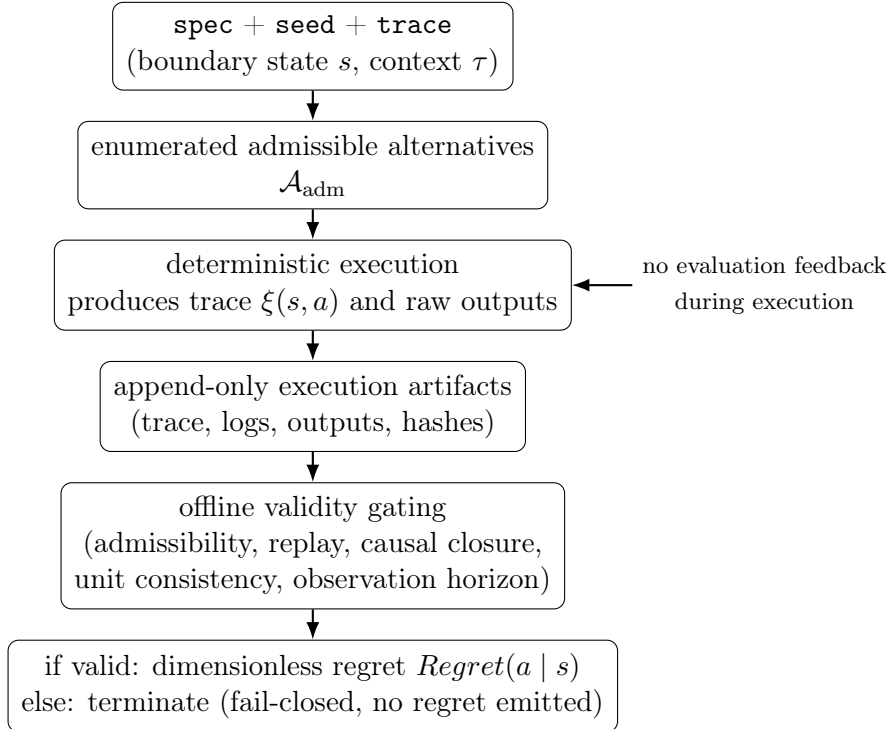


Figure 1. Evaluation pipeline under deterministic replay. Execution produces append-only artifacts. Validity checks gate regret emission; failures terminate without emitting regret.

3 Regret

Regret is counterfactual loss relative to the best admissible alternative, evaluated under identical boundary conditions.

3.1 Scalar Regret (Single Metric)

Let $J(s, a)$ be a scalar objective to maximize. Let a^* be the best admissible alternative as defined in Section 3.3.

$$\text{Regret}(a \mid s) = J(s, a^*) - J(s, a) \quad (2)$$

3.2 Multi-Metric Regret Requires Nondimensionalization

In engineering systems, metrics have different physical dimensions (latency, throughput, energy, error rate). Direct weighted summation of heterogeneous units is dimensionally invalid and produces scale bias.

Define metric components:

- $M_i(s, a)$ for $i = 1, \dots, k$, each with declared physical units and an associated characteristic scale $C_i(s)$ sharing the same units.

Define a **characteristic scale** $C_i(s)$ with the same units as M_i . Both the units and the scale definition are mandatory inputs to the evaluation and must be specified prior to execution. The normalization transform is:

$$r_i(s, a) = \frac{M_i(s, a^*) - M_i(s, a)}{C_i(s)} \quad (3)$$

Then define dimensionless aggregate regret:

$$\text{Regret}(a \mid s) = \sum_{i=1}^k w_i r_i(s, a) \quad (\text{defined only if all } r_i \text{ are dimensionless}). \quad (4)$$

Constraints.

- $w_i \geq 0$
- $\sum_i w_i = 1$ is recommended for interpretability.

ASCII form.

```
r_i(s,a) = (M_i(s,a*) - M_i(s,a)) / C_i(s)
Regret(a|s) = sum_i w_i * r_i(s,a)
```

3.2.1 Recommended Choices for $C_i(s)$

Pick one and document it per metric:

1. Range scale (robust default)

$$C_i(s) = \max_{a \in A_{\text{adm}}} M_i(s, a) - \min_{a \in A_{\text{adm}}} M_i(s, a) \quad (5)$$

2. Tolerance scale (SLO/SLA anchored)

$$C_i(s) = \text{SLO}_i \quad (6)$$

(for “must not exceed” constraints)

3. Utopia (fraction-of-optimum, use with care)

$$C_i(s) = |M_i(s, a^*)| \quad (7)$$

(singular when optimum is 0)

This mirrors nondimensionalization practice in physics codes where equations are commonly expressed in nondimensional form to preserve invariance under unit choice.

3.3 Comparator Definition: Best Admissible Alternative

The comparator must be **admissible**: it cannot depend on information unavailable to the evaluated agent at decision time.

Let $I(h)$ be the information set induced by observable history h , parameterized by an explicit observation horizon that defines which variables, time steps, and signals are visible.

An alternative a is admissible if it maps a declared information set $I(h)$ to decisions without using hidden state (private variables, future RNG outcomes, or external oracle information), and if its declared observation horizon does not exceed that of the evaluated agent.

$$a \in A_{\text{adm}} \iff \text{Dep}(a) \subseteq I(h) \quad (8)$$

Then:

$$a^* = \arg \max_{a \in A_{\text{adm}}} J(s, a) \quad \text{subject to } \mathcal{O}(a) \preceq \mathcal{O}(a_{\text{eval}}) \quad (9)$$

This prevents “clairvoyance regret”: penalizing agents for not using information outside their declared observation horizon. Such comparisons are invalid counterfactuals and are treated as correctness failures.

4 Scalarization Limits and Pareto Structure

Weighted-sum scalarization induces a total order over alternatives and therefore selects points on the Pareto frontier. In non-convex tradeoff surfaces, some Pareto-optimal alternatives are not representable by any fixed weight vector.

This is a known limitation of linear scalarization. In this framework, scalarization does not resolve Pareto incomparability; it preserves it by construction and may leave some alternatives unordered under the induced comparison.

If non-convex Pareto coverage is required, an alternative scalarization (e.g., Chebyshev or max-norm formulations) may be used, but it must be treated as a distinct objective definition. Changing the scalarization defines a new task context τ and invalidates cross-context comparison.

Optionally, the set of non-dominated alternatives (the Pareto set) may be emitted as an evaluation artifact alongside any scalarized regret values.

5 Conservation Laws (Correctness Invariants)

5.1 Conservation of History (Deterministic Replay)

With fixed (`spec`, `seed`, `trace`) and complete state s , the execution is a pure function and produces bitwise-identical outputs across replays, including:

- identical derived metrics
- identical decision sequence
- identical derived-state hash

Any divergence indicates entropy injection (e.g., race conditions, nondeterministic iteration order, unseeded randomness, external time sources, or network I/O) and constitutes a correctness failure.

5.2 Conservation of Trace (Boundary Condition Invariance)

The workload trace is immutable and independent of the alternative being evaluated.

If choosing an alternative a changes the subsequent trace arrival pattern and that feedback is not modeled deterministically within the frozen system state s , the counterfactual comparison is invalid. Such cases violate causal closure and render the counterfactual undefined.

5.3 Conservation of Dimensionality

All aggregated regret scalars must be dimensionless.

Implementations must reject any configuration that aggregates metrics with missing unit annotations, missing characteristic scales, or heterogeneous units without explicit $C_i(s)$ normalization. Such violations are treated as correctness failures and must abort evaluation.

5.4 Offline Isolation

Metric computation and regret attribution occur strictly post-execution.

No gradient updates, parameter changes, prompt edits, policy modifications, or online learning operations occur during measurement.

6 Validity Checks and Hard Failures

The following conditions are treated as correctness failures, not as large or undesirable regret values after admissible maximization:

- **Trace divergence:** divergence between traces produced by admissible alternatives, indicating invalid counterfactual evaluation.
- **Unit or scale violation:** missing unit annotations, missing characteristic scales, or aggregation across heterogeneous units without normalization, indicating dimensional inconsistency.
- **Oracle violation:** use of a comparator whose declared observation horizon strictly exceeds that of the evaluated agent, indicating inadmissible access to hidden or future information.

If any correctness failure occurs, the evaluation must fail closed and must not emit regret values.

7 Artifact Mapping

Theoretical objects are mapped to concrete, append-only artifacts as follows:

- s (boundary conditions): **traces/** plus frozen environment state.
- A and admissibility constraints: **specs/** (action sets, policies, declared observation model, and observation horizon).
- M_i : replay-derived metrics recorded in **runs/** or **traces/** as append-only outputs.
- C_i : characteristic scales defined in specifications and versioned with the evaluation.
- Regret: evaluation artifact emitted by the offline analyzer.

All artifacts are append-only and versioned; modification or mutation invalidates the evaluation.

8 Minimal Falsification Scenario (Asymmetric Information Externality)

A minimal scenario that falsifies naive regret-evaluation engines consists of:

- two agents,
- a single decision step,
- two actions,
- asymmetric private information,
- an externality cost.

Naive engines define a^* using hidden state and therefore measure “luck regret.”

Correct engines define a^* over admissible alternatives only and measure “decision regret.”

This scenario violates admissibility by constructing a^* from undeclared information and therefore breaks Conservation of History and oracle admissibility. It should be implemented as a unit test or CI gate for any regret-evaluation engine claiming mechanical validity.